

24 Lecture 24: Quantum search algorithm (Grover)

24.1 The problem

The problem is to find, in a set of $N = 2^n$ elements, a subset of M elements that satisfies some given conditions. We say then that the search problem has M solutions.

We can assume the existence of a function $f(x)$ from n bits to 1 bit, that takes the value $f(x) = 1$ if x is a solution and $f(x) = 0$ if x is not a solution.

We also assume that there is a black box (an “oracle”) U_f able to execute the operation on the $n + 1$ qubit state $|x\rangle|q\rangle$, where $|x\rangle$ is a n -qubit state and $|q\rangle$ is a 1-qubit state:

$$U_f|x\rangle|q\rangle = |x\rangle|q \oplus f(x)\rangle \quad (24.1)$$

In particular

$$U_f|x\rangle|0\rangle = |x\rangle|f(x)\rangle \quad (24.2)$$

24.2 A two-bit example

In this example $N = 4$, and the oracle that tests x is one of the four gates:

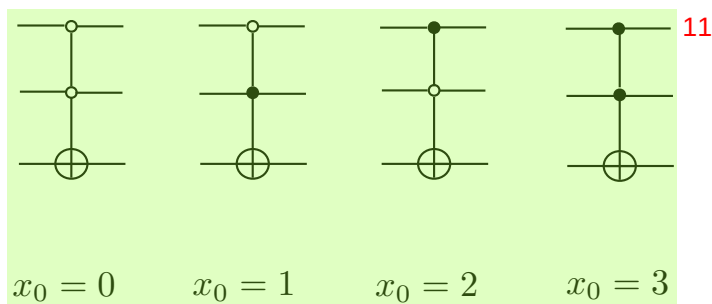


Fig. 24.4 Oracles for the four cases: solution $x_0 = 0$, ... solution $x_0 = 3$.

each corresponding to a particular solution x_0 .

The first two qubits (the “query” qubits) encode x , the last qubit is the oracle response. The circuit for the search of the solution is:

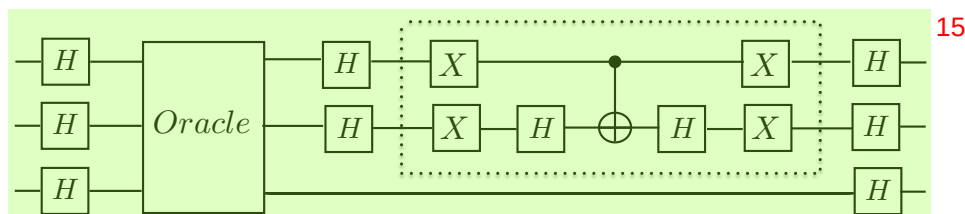


Fig. 24.5 circuit for the search of the solution x_0

Initially, the two query qubits are in the state $|00\rangle$, and the last qubit in the state $|1\rangle$. Only one iteration is required to obtain exactly x_0 . One can verify, using the circuit, that the measurement of the top two qubits gives x_0 after using the oracle only once. 1

By contrast, a classical computer trying to find x_0 by examining all the two-bit x (four of them) would require an average of 2.25 oracle calls. 2

Exercise: prove it. 3